

Underwater Vision-Based Surveillance Using Machine Learning and Artificial Intelligence

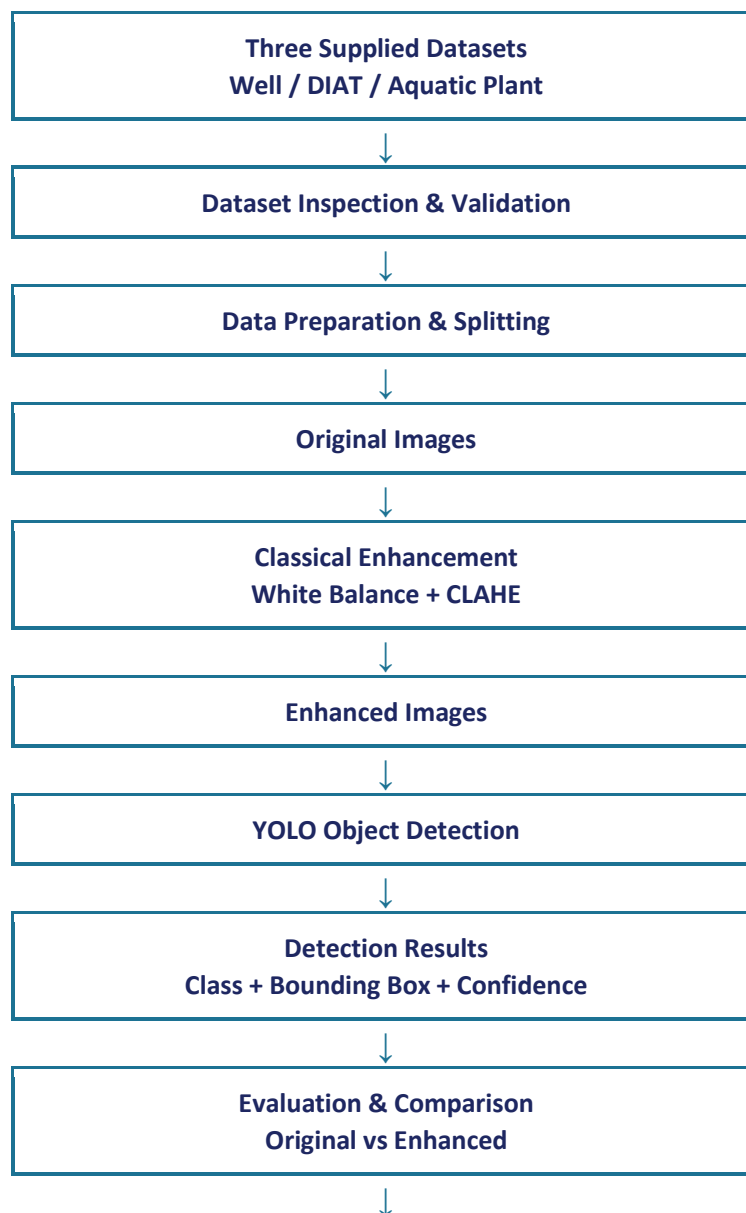
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Day 5: PIPELINE & EXPERIMENT PLANNING

1. Objective

The objective of Day 5 is to design a reproducible workflow from the supplied datasets to final evaluation, define the baseline, advanced methods and project-specific experiment, identify evaluation criteria, and prepare the experiment log and repository structure.

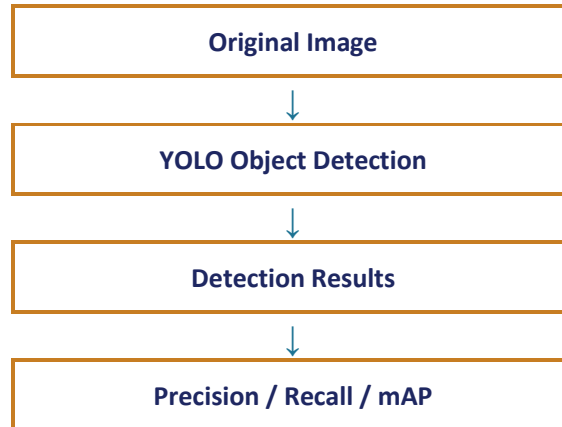
2. Pipeline — Flowchart



3. Baseline Method

Baseline Method — Original Image → YOLO

Purpose: Establish the reference detection performance without any image enhancement.



This baseline will be used as the reference for comparing the enhanced-image experiments.

4. Experiment 1 — Classical Enhancement

Classical Image Enhancement

Purpose: Improve underwater image quality using traditional image-processing techniques.

Methods

- White Balance
- CLAHE

Process

Original Image → White Balance + CLAHE → Enhanced Image

The enhanced images will be evaluated using suitable image-quality measures such as PSNR, SSIM and UIQM where applicable.

5. Experiment 2 — YOLO Object Detection

Purpose: Determine whether classical image enhancement improves object detection.

Two conditions will be compared:

Original Image → YOLO → Detection Results

vs.

Classically Enhanced Image → YOLO → Detection Results

Evaluation will use

- Precision
- Recall
- mAP@0.5
- mAP@0.5:0.95
- processing time

6. Experiment 3 — Advanced Method

Advanced Method — Water-Net / FUnIE-GAN

Purpose: Evaluate learning-based underwater image enhancement methods.

Candidate methods

- Water-Net
- FUnIE-GAN

Process

Original Image → Water-Net / FUnIE-GAN → Enhanced Image → YOLO → Detection Results

The results will be compared with the baseline and classical enhancement results.

7. Evaluation Criteria

Object Detection Metrics

- Precision
- Recall
- mAP@0.5
- mAP@0.5:0.95
- Inference / processing time

Image Enhancement Metrics

- PSNR
- SSIM
- UIQM

These will be used where applicable.

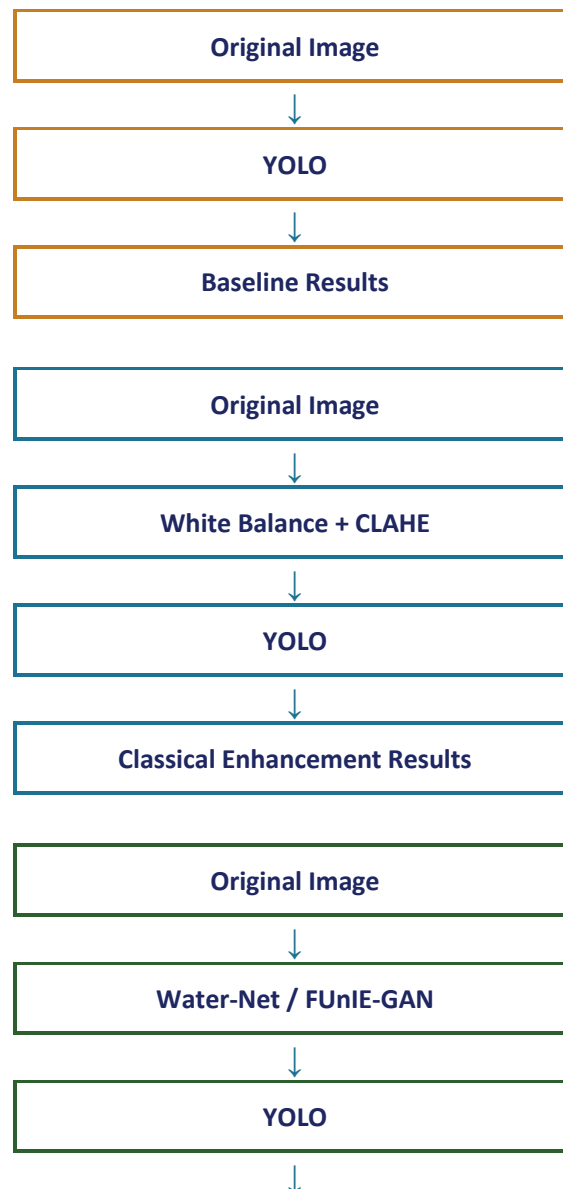
Tracking

Not evaluated now due to no video data.

8. Project-Specific Experiment

Original vs. Classical vs. Learning-Based Enhancement

The project will compare object detection performance under three conditions:



10. Repository / Folder Structure

```
Underwater-vision-surveillance/  
├── datasets/  
│   ├── well/  
│   ├── diat_aquarium/  
│   └── aquatic_plant/  
├── preprocessing/  
│   ├── white_balance/  
│   └── clahe/  
├── enhancement/  
│   ├── water_net/  
│   └── funie_gan/  
├── detection/  
│   └── yolo/  
├── experiments/  
│   ├── baseline/  
│   ├── classical_enhancement/  
│   └── learning_based_enhancement/  
├── evaluation/  
├── results/  
├── logs/  
│   └── experiment_log.csv  
├── notebooks/  
├── docs/  
└── README.md
```